Quantum States as Control Variables in Gravity Tests: A Unified Zero-Background Theorem

MuningAn · PlanetarySystem · 2026-08 · Paper E5 (first version, with explicit falsification clause)

Abstract Gravity tests have traditionally treated quantum states as "cleaner detectors"; this paper shows that quantum states can do more—they can serve as **control variables**. We unify three scheduled state-control tests (the nuclear-clock state-dependence search, GIE internal-state control, and spin-resolved frame dragging) under one theorem: in a local, Lorentz-invariant, weakly coupled effective field theory (EFT), the dependence of the gravitational/Yukawa coupling on the probe's internal state is suppressed by dimension-6 power counting—$\delta\frac{\alpha}{\alpha} = \angle O_{\text{eff}} \angle \frac{M_{\text{Pl}}}{\sqrt{\alpha}\Lambda^2 m_{\text{probe}}}$ for a massive mediator, and $\delta\frac{G}{G} = \angle O \angle \frac{M_{\text{Pl}}^2}{\Lambda^2 d^2 M_{\text{src}} m_{\text{probe}}}$ for the massless graviton. The EFT upper bounds of the three cases lie 10.7, 13.1, and 28.3 orders of magnitude below their respective readout sensitivities: **state-control tests are zero-background within the entire class of weakly coupled local EFTs**. Any observable state-dependent signal therefore simultaneously falsifies general relativity, standard KK, and every weakly coupled local EFT, pointing to strongly coupled, non-local, or state-charged gravitational sectors. The theorem carries an explicit falsification clause and a "state-control column" for the adjudication matrix. All numbers come from experiments/18; Fig. 8 visualizes the gaps.**Keywords** quantum states; gravity tests; effective field theory; zero background; state-dependent coupling; adjudication matrix

Introduction: From "Which Detector" to "Which State"

The experimental history of gravity testing is a history of detectors: torsion balances, atom interferometers, nuclear clocks. Each step reduces systematic errors, but the detector always plays a passive role—it measures gravitational effects in a given geometry. This paper proposes a different use: **the quantum state itself as a control variable**. The probe's internal state (nuclear isomer state, electronic internal state, spin orientation) becomes a switchable experimental knob; the presence or absence of a state-dependent signal is itself the object of adjudication.

This idea exists in the literature in scattered form: the advanced stage of the nuclear-clock design proposes microwave control of the nuclear internal state; the upgraded GIE proposal asks whether the entanglement coupling is modulated by the quantum state. This paper unifies them into one framework and gives a unified zero-background theorem.

General Framework: Four Ingredients and Three State-Control Channels

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A gravity test consists of four ingredients: source (mass distribution), probe (quantum system), observable (phase/frequency/entanglement), and state (control over the probe's internal state). Traditional experiments optimize the first three; state-control experiments make the fourth an independent variable. There are three state-control channels:

- **Internal-state channel**: two internal states of a single system (ground/isomer states of the nuclear clock)—the distinguishing operator is the nuclear quadrupole moment Q_{ij};
- **Superposition channel**: mass superposition and entanglement as a resource (GIE)—the internal-state-control version uses the electronic internal state (dipole/spin) as the distinguisher;
- **Spin-orbit channel**: spin orientation as the knob (frame-dragging readout)—the distinguisher is the spin S.

The common structure of the three channels: the matrix element $\angle O \angle$ of the state-distinguishing operator O sets the theoretical ceiling on the state-dependent coupling.

Theorem: The Universal EFT Bound on State-Dependent Coupling

Theorem (Zero-background property of state control). In a local, Lorentz-invariant, weakly coupled EFT (cutoff Λ), let the probe (mass m_{probe}) have internal states distinguished by the operator O, with Wilson coefficient $c \sim O(1)$.

(1) **Massive mediator** (strength α relative to gravity, range λ): the leading state-distinguishing realization is the dimension-6 operator $cO_{\text{ij}}\partial_i\partial_j\frac{\varphi}{\Lambda^2}$; in the static limit,

Test	Distinguishing operator	EFT upper bound	Readout sensitivity	Zero-background gap	Script/Fig.
Nuclear-clock state dependence	$Q \sim 9.4$ b	$\delta\alpha = 5.4 \times 10^{-22}$	3×10^{-11} (5σ)	10.7 orders	17, Fig. 7
GIE internal-state control	$e \cdot a_{\mathrm{B}}$ (conservative)	$\delta\frac{G}{G} = 7.4 \times 10^{-16}$	10^{-2} (phase)	13.1 orders	18, Fig. 8
Frame dragging	$S \sim 1$	$\delta\frac{G}{G} = 4.9 \times 10^{-31}$	10^{-2} (phase)	28.3 orders	18, Fig. 8

Table 1: Table 1: The three cases. All at $\Lambda = 10$ TeV, $c = 1$; GIE at $m = 10^{-14}$ kg, $d = 450\mu$m; nuclear clock at $\alpha = 10^{-6}$, $\lambda = 1$ m. The bounds tighten as Λ^2; the gaps only grow.

$$\frac{\delta\alpha}{\alpha} = \frac{c\angle O_{\mathrm{eff}}\angle M_{\mathrm{Pl}}}{\sqrt{\alpha}\Lambda^2 m_{\mathrm{probe}}}, \quad \angle O_{\mathrm{eff}}\angle = \angle O\frac{\angle}{\lambda^2}. \tag{1}$$

(2) **Massless graviton** (source mass M_{src}, distance d): the ratio of the state-dependent potential to the gravitational potential is

$$\frac{\delta G}{G} = \frac{c\angle O\angle M_{\mathrm{Pl}}^2}{\Lambda^2 d^2 M_{\mathrm{src}} m_{\mathrm{probe}}}. \tag{2}$$

Corollary (zero background): Both upper bounds lie more than 10 orders of magnitude below the readout sensitivities of all scheduled state-control experiments (Table 1, Section 4). Hence **state-control tests expect zero background within the class of weakly coupled local EFTs**; any observable state-dependent signal constitutes a simultaneous falsification of GR, standard KK, and all weakly coupled local EFTs.

Eq. Equation 1 generalizes Corollary 4 of the companion classification theorem (Paper T2): Corollary 4 is the special case $O = Q_{\mathrm{ij}}$, $m_{\mathrm{probe}} = m_{\mathrm{Th}}$; Eq. Equation 2 covers the state dependence of the graviton itself (the state-resolved version of the equivalence principle).

Three Cases

The ordering of the gaps (nuclear clock $<$ GIE $<$ frame dragging) reflects differences in probe mass and distance, not in methodology—the common point of the three is that **all gaps are enormous**. The conclusion is robust to the choice of distinguishing operator (for GIE, taking the NV spin magnetic moment instead of the atomic dipole raises the gap to 31 orders; see script 18).

The State-Control Column of the Adjudication Matrix

The methodological discipline of the adjudication matrix is one discriminating channel per row. The state-control framework adds a column: whenever a probe possesses an internal-state distinction, that row automatically acquires a zero-background state-control test. Concretely:

- Row 12 (KK extra dimensions): the nuclear-clock state-dependence search (Corollary 4);
- Rows 6-7 (GIE): the internal-state-control upgrade;
- Row 8 (frame dragging): spin-resolved readout.

The three columns share one zero-background theorem—the state-control framework's payback to the matrix methodology: the falsifiability of a test is upgraded from "search within a window" to "zero-background falsification".

Falsification Clause

The theorem's falsification clause agrees with the fifth clause of Paper T2 and is refined here: **if any state-control experiment measures a state-dependent signal above the Table 1 readout sensitivity, then GR, standard KK, and every weakly coupled local EFT are simultaneously falsified**, and the new physics must belong to one of three classes: a strongly coupled sector (cutoff at the ≤ 43 MeV scale, see script 17), a non-local theory, or a state-charged gravitational sector (gravitational coupling carrying state quantum numbers). The testable signatures of the three—the energy structure of strong coupling, propagator modifications of non-locality, and the selection rules of state charge—constitute the search list for follow-up work.

Conclusion

The quantum state is the fourth variable of gravity testing. This paper gives its unified framework and zero-background theorem: three channels (internal-state / superposition / spin-orbit) share one dimension-6 power counting, and the zero-background gaps of the three scheduled experiments range from 10.7 to 28.3 orders of magnitude. State-control tests are thus "zero-background falsification" experiments: with no signal they confirm the self-consistency of the EFT picture; with a signal they open, in one stroke, three exits—strong coupling, non-locality, or state charge. This is the upgrade from "measuring gravity" to "measuring the state dependence of gravity".

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Code availability experiments/17 (Corollary 4), 18 (the unified zero-background theorem), and Figs. 7-8 at github.com/everest-an/Antigravity.